10-Mark Questions & Answers MPCA Unit 3: The Intel Microprocessor 8086 Architecture

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Q1. Explain the internal architecture of 8086 in detail (BIU, EU, registers, flag register) with a neat labeled diagram.

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1. Bus Interface Unit (BIU)

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2. Execution Unit (EU)

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Q2. Explain the functional pin diagram of 8086 microprocessor in detail, describing the function of each pin

group (address/data bus, control signals, power supply).

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1. **Multiplexed Address/Data Bus ($ ext{AD}_0 - ext{AD}_{15}$):** Time-multiplexed lines carrying lower 16-bit memory address during $T_1$ and 16-bit data during $T_2-T_4$.

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**Q3. Explain memory segmentation and banking in 8086 in detail with physical address calculation examples. Also explain why segmentation is used.**

#### 1. Why Memory Segmentation is Used

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#### 2. Physical Address Calculation Formula & Example

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- **Example 1 (Instruction Fetch):** `CS = 3000H`, `IP = 0100H`

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3. Memory Banking System (Even & Odd Banks)

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1MB P
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Q4. Explain the functioning of 8086 in both minimum mode and maximum mode with neat diagrams, and state the key differences.

1. Minimum Mode Block Diagram ($ ext{MN}/\overline{ ext{MX}} = 1$)

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2. Maximum Mode Block Diagram ($ ext{MN}/\overline{ ext{MX}} = 0$)

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Q5. Draw and explain in detail the timing diagrams for read and write operations in both minimum and maximum mode of 8086.

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MINIM
CLK    : __--__--__--__--__--__--__--__--__--
|   T1
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Q6. Explain the interrupt structure of 8086 in detail, including types of interrupts (hardware/software), interrupt vector table, and interrupt servicing procedure.

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INTER
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Q7. Explain the process of demultiplexing the address/data bus in 8086 with a circuit diagram (using latch 74LS373).

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